

RAG Chatbot in n8n - Complete Build Guide

Updated stack: n8n Cloud | Pinecone (1536) | OpenAI embeddings | Gemini chat model

1. What you are building

A chatbot that answers questions using only your own PDF documents (RAG = Retrieval Augmented Generation). It uses two n8n workflows:

- **Workflow A (Ingestion):** upload a PDF -> split into chunks -> convert to embeddings -> store in Pinecone.
- **Workflow B (Chat):** user question -> AI Agent searches Pinecone -> answers using only what it found.

```
WORKFLOW A
  Form Trigger -> Pinecone Vector Store (Insert)
    + Embeddings OpenAI
    + Default Data Loader <- Text Splitter

WORKFLOW B
  Chat Trigger -> AI Agent
    + Chat Model (Gemini)
    + Simple Memory
    + Pinecone Vector Store (Tool) <- Embeddings OpenAI
```

2. Final tool choices

Part	Choice
Automation platform	n8n Cloud (free trial)
Vector database	Pinecone Starter plan. Index name: rag-docs, Vector type: Dense, Dimension: 1536, Metric: cosine
Embeddings (both workflows)	OpenAI text-embedding-3-small (outputs 1536 dimensions)
Chat model (Workflow B)	Google Gemini Chat Model (pick a current Flash model in the dropdown)
Memory	Simple Memory (window buffer)
Documents	TechNova_Employee_Handbook.pdf, TechNova_IT_Security_Policy.pdf, NovaDesk_Customer_FAQ.pdf
Test sheet	Google Sheet "RAG CHATBOT" with 20 questions (see Appendix)

Two golden rules

1. The embedding model must be the **same** in Workflow A and Workflow B (Embeddings OpenAI, text-embedding-3-small).
2. The Pinecone index dimension must **match** the embedding model output (1536). The dimension cannot be changed after creating the index; if wrong, delete the index and create a new one.

Do not use Gemini embeddings with this index: Gemini embedding models output a different size (Google deprecated text-embedding-004, and gemini-embedding-001 outputs 3072 by default).

3. Accounts and keys (do this first)

- 1 **OpenAI:** create an API key at platform.openai.com (needed for embeddings). Check whether n8n Cloud offers free OpenAI credits on the credential screen.

- 2 **Google Gemini:** create an API key at aistudio.google.com (Get API key -> Create API key), or use free credits if n8n offers them for the Gemini node.
- 3 **Pinecone:** sign up at pinecone.io (free Starter plan) and copy your API key from the API Keys page.
- 4 **Telegram (later, optional):** message @BotFather, send /newbot, and copy the bot token.

4. Create the Pinecone index

- 1 In Pinecone click **Create index**.
- 2 Name: **rag-docs**.
- 3 Tick **Custom settings**.
- 4 Vector type: **Dense**.
- 5 Dimension: **1536** (do not leave it empty).
- 6 Metric: **cosine**.
- 7 Keep the default cloud and region, then click **Create index**.

5. Add credentials in n8n

Open **Credentials** -> **Create credential** and add three: **Pinecone API**, **OpenAI API**, and **Google Gemini (PaLM) API**. Paste the matching key in each and save.

6. Workflow A: Ingestion (name it "RAG - Ingestion")

Node 1: n8n Form Trigger

- 1 Click **Add first step** and search **n8n Form**.
- 2 Under **Triggers** choose **On new n8n Form event** (lightning bolt icon). The node is then titled "On form submission".
- 3 Form Title: **Upload Document** (required).
- 4 Click **Add Form Element**: Label **Document** (exact spelling), Element Type **File**, Required on. Under Add Attributes set Accepted File Types to **.pdf**.
- 5 Leave Authentication as None. Click Back to canvas.

Common mistake: do not choose "Next Form Page" or "Form Ending" (they are under Actions). If you see a "Page Type" dropdown and a yellow warning saying a Form Trigger must be set up first, you picked the wrong node. Delete it and add the trigger.

Node 2: Pinecone Vector Store (insert)

- 1 Click the + after the trigger, search **Pinecone Vector Store**, choose the action **Add documents to vector store** (Insert Documents mode).
- 2 Credential: your Pinecone credential. Pinecone Index: **rag-docs**.
- 3 Optional: Options -> Pinecone Namespace (for example handbook). If you use a namespace here, use the identical namespace in Workflow B.
- 4 Leave Clear Namespace off (turn it on only if you want to wipe old vectors before an upload).

Node 3: Embeddings OpenAI (sub-node)

- 1 Click the **Embedding** + under the Pinecone node and choose **Embeddings OpenAI**.
- 2 Credential: your OpenAI credential. Model: **text-embedding-3-small**.

Node 4: Default Data Loader (sub-node)

- 1 Click the **Document** + and choose **Default Data Loader**.
- 2 Type of Data: **Binary**. Data Format: **Automatically Detect by Mime Type**.
- 3 Text Splitting: **Custom** (this opens the Text Splitter slot).
- 4 Options -> Add Metadata: name **source**, value **{{ \$binary.Document.fileName }}** (the word Document must match your form label).

Node 5: Recursive Character Text Splitter (sub-node)

- 1 Click the **Text Splitter** slot and choose **Recursive Character Text Splitter**.
- 2 Chunk Size: **1000**. Chunk Overlap: **200**.

Run and verify

- 1 Click **Execute Workflow**. A test form page opens. Upload the first PDF and submit.
- 2 Open Pinecone -> index rag-docs. The record count must go up.
- 3 Repeat for the other two PDFs (run Execute Workflow again each time).
- 4 If the count stays at 0, read the error on the Pinecone node. The usual cause is a dimension mismatch.

7. Workflow B: Chat (name it "RAG - Chatbot")

Node 1: Chat Trigger

Add **When chat message received**. Keep the defaults.

Node 2: AI Agent

- 1 Add **AI Agent** and connect it to the Chat Trigger.
- 2 Source for Prompt: **Connected Chat Trigger Node**.
- 3 Options -> System Message: paste the text below.

```
You are a helpful assistant that answers questions about our documents.  
Rules:  
1. ALWAYS use the knowledge base tool to search before answering.  
2. Answer ONLY from the retrieved information.  
3. If the answer is not in the retrieved information, reply exactly:  
   "I couldn't find that in the documents."  
4. Keep answers short and clear. Mention the source file when possible.
```

Node 3: Chat Model

- 1 Under **Chat Model** choose **Google Gemini Chat Model**.
- 2 Credential: Gemini. Model: a current Flash model from the dropdown.
- 3 Options -> Temperature: **0.2** (lower means fewer invented answers).

Node 4: Simple Memory

- 1 Under **Memory** choose **Simple Memory**.

- 2 Session ID: from Chat Trigger (default). Context Window Length: 5.

Node 5: Pinecone Vector Store as a tool

- 1 Under **Tool** choose **Pinecone Vector Store**.
- 2 Operation Mode: **Retrieve Documents (As Tool for AI Agent)**.
- 3 Credential, Index (rag-docs) and Namespace: same as Workflow A.
- 4 Name: **knowledge_base**.
- 5 Description: **Searches the uploaded company documents. Use it for any question about policies, procedures, pricing, or facts in the documents.**
- 6 Limit: **4**. Include Metadata: on.

Node 6: Embeddings for retrieval

In the Pinecone tool Embedding slot add **Embeddings OpenAI** with model **text-embedding-3-small** (identical to Workflow A).

8. Test the chatbot

- 1 Click **Open Chat** at the bottom of the canvas.
- 2 Ask the 20 questions from the Appendix and fill in the Bot answer and Pass/Fail columns of your sheet.
- 3 Click the AI Agent node after a run to confirm it called the knowledge_base tool and see which chunks came back.
- 4 Accuracy formula for the sheet: **=COUNTIF(G2:G21,"Pass")/20** (format the cell as percentage).
- 5 For out-of-scope questions the bot passes only if it declines. If it answers from general knowledge, mark Fail and tighten the system prompt.

9. Telegram version (optional)

Duplicate Workflow B and name it "RAG - Telegram". Keep the original for testing.

- 1 In Telegram message @BotFather, send /newbot, and copy the token. Add a Telegram credential in n8n.
- 2 Delete the Chat Trigger and add **Telegram Trigger** (Updates: message).
- 3 AI Agent: Source for Prompt = Define below, Text = **{{ \$json.message.text }}**.
- 4 Simple Memory: Session ID = Define below, Key = **{{ \$json.message.chat.id }}**.
- 5 After the agent add **Telegram -> Send a text message**. Chat ID = **{{ \$('Telegram Trigger').item.json.message.chat.id }}**, Text = **{{ \$json.output }}**.
- 6 Publish (activate) the workflow and message your bot. Deactivate it when you are not demoing to save executions.

10. Upgrades (pick 2 or 3)

Upgrade	How
Auto-ingest	Replace the Form Trigger with a Google Drive Trigger (file created in folder) + Google Drive Download File.

Upgrade	How
Error alerts	Create a workflow starting with Error Trigger that sends a Telegram alert. Then set it in each workflow under Settings -> Error Workflow.
Duplicate protection	Keep a Google Sheet of ingested file names and check it with an IF node before inserting.
Better retrieval	Test chunk sizes 800/1000/1200 and Limit 3/4/6. Record which scores best.
Source citations	Add to the system prompt: end each answer with the source file name.

11. Package it for your resume

- 1 Export both workflows as JSON (top-right menu -> Download) **before the trial ends**.
- 2 Create a GitHub repo with: README (problem, architecture diagram, tools, setup steps, test results), /workflows JSON files, /screenshots.
- 3 Put your test table and accuracy score in the README.
- 4 Record a 1-2 minute demo video: upload a PDF, ask questions, ask an out-of-scope question, show Telegram.

Sample resume bullet (replace numbers with your real results):

Built a RAG chatbot in n8n using OpenAI embeddings, Pinecone and Gemini that answers questions from uploaded PDFs via Telegram; scored 17/20 on a custom test set and correctly declined out-of-scope questions.

12. Troubleshooting

Problem	Likely cause and fix
Insert fails, 0 records	Dimension mismatch. Index must be 1536 and Embeddings OpenAI must use text-embedding-3-small.
Yellow warning: Form Trigger must be set up first	You added Next Form Page. Delete it and add On new n8n Form event.
Bot answers without searching	Tool description too vague, or the system prompt does not require the tool.
Irrelevant or wrong answers	Different embedding model in A and B, or different namespace.
Misses details in the PDF	Increase Limit; try chunk size 800 to 1200; make sure the PDF has real text (scanned images need OCR).
Telegram bot silent	Workflow not published/active, or wrong credential.
Running out of executions	Test with small PDFs and deactivate the Telegram workflow when not demoing.

Appendix: 20 test questions

Do not upload this guide to Pinecone. It is for you only.

#	Question	Expected answer	Source
1	How many casual leaves per year?	14 days	Handbook
2	What are the regular working hours?	10:30 AM to 7:00 PM, Monday to Friday	Handbook
3	How long is the probation period?	5 months	Handbook
4	Which day is the mandatory in-office day?	Wednesday	Handbook
5	Password length and change frequency?	At least 14 characters, every 75 days	IT Policy

#	Question	Expected answer	Source
6	After how many minutes does the screen auto-lock?	4 minutes	IT Policy
7	NovaDesk refund window?	9 days from first payment	FAQ
8	Free trial length and credit card needed?	21 days, no credit card	FAQ
9	What does the Growth plan cost?	Rs. 1,899 per agent per month	FAQ
10	I join on July 10. Casual leaves this year?	6	Handbook
11	11 unused earned leaves at year end. How many carry forward?	7	Handbook
12	Annual cost of Growth plan for 4 agents?	Rs. 75,960 (1,899 x 4 x 10 months)	FAQ
13	Notice period during probation vs after confirmation?	15 days vs 45 days	Handbook
14	Laptop stolen: who to contact and by when?	IT helpdesk within 2 hours, then tell manager	IT Policy
15	Work from home days and what to do for internal tools?	Up to 2 days per week with approval; use NovaSecure VPN	Handbook + IT
16	HR email and IT helpdesk hours?	hr@technova.example; 8:00 AM to 9:00 PM, Mon to Sat	Handbook + IT
17	What is the CEO salary?	I couldn't find that in the documents	none
18	Does TechNova offer stock options?	I couldn't find that in the documents	none
19	What is the capital of France?	I couldn't find that in the documents	none
20	(a) How many sick leaves? (b) Do I need a certificate for that?	(a) 9 days per year; (b) only for 3 or more consecutive days	Handbook